

Sample Report on the User-Centered Design Process Used for Voting Systems

This document provides a sample report on the user-centered design (UCD) process to meet the requirements under Principle 2.2 as part of the certification of a voting system to the Voluntary Voting System Guidelines (VVSG) 2.0.

It is one of ten companion documents to the NIST VTS 400-5 Handbook for VVSG 2.0 Usability and Accessibility Test Strategies and can be downloaded with the Handbook. All documents are also available on the NIST Voting Program website at <https://www.nist.gov/itl/voting/vts-400-5-documents>.

Any content provided in this document is just a sample to show the type of information to include in any section.

User-Centered Design Report for
ABC Voting System
for VVSG 2.0 Requirement 2.2-A

ABC Elections 1.0

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1.0 Overview

This report describes the user-centered design (UCD) process for ABC Elections Voting System version 1.0 conducted during the design and development process to meet VVSG 2.0 Requirement 2.2-A.

The scope of this work includes the design and development of the ABC Voting System ballot marking device.

1.1 UCD Scope and Participants

From January 1, 2022 – December 31, 2022, the UCD process included a variety of usability tests and other UCD methods, including:

1. 4 usability tests with a total of 86 sessions
2. 1 prototyping session with 9 participants
3. 1 expert review with 5 accessibility experts
4. 36 interviews and observations with voters and poll workers

Participants in these activities included:

1. 90 total voters, including:
 2. [##] blind voters using the audio format and tactile keypad
 3. [##] voters with low vision using audio format with visual display
 4. [##] voters with low vision using screen text size and contrast
 5. [##] deaf or hard of hearing voters
 6. [##] voters with mobility disabilities
 7. [##] voters with dexterity disabilities using the tactile keypad
 8. [##] voters with dexterity disabilities using a dual switch
 9. [##] voters with low literacy
 10. [##] voters using alternative languages
11. 18 total poll workers or other election workers

1.2 Impact

Many design decisions were made as a result of the user-centered design activities. They included:

1. Designing the initiation of a voting session so blind voters can complete the activities themselves by changing the planned position of the audio headset jack and adding an audio message about how to start the session.
2. Adding information to the audio format to help voters navigate more accurately and quickly.
3. Changing the design of the keys on the tactile keypad to be more distinct and adjusting the Braille labels to more accurately identify each key.
4. [And so on ...]

2.0 Product Description

2.1 Full Product Description

This report is for ABC Elections Voting System version 1.0. Designed to present ballots to voters throughout the U.S. and collect voter responses, it consists of a touch screen tablet, a connected COTS printer, and optional accessibility devices including a 7-button tactile keypad, 2-button dual switch, and audio output.

[There can be additional details such as details of the components, software, and illustrations of the voting system components.]

3.0 Timeline

Timeline and summary of UCD work on ABC Voting 1.0

4.0 User-Centered Design Activity Detail

4.1 Interviews with voters

About this activity

Activity process and results detail